Summations, geometric and arithmetic progressions

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Outline

Summations

Geometric progressions

Arithmetic progressions

Summations

Introduction: Summations

- In this section we introduce the **summation symbol** $\sum$, a compact notation to write sums of many terms
- We will see its **six main properties** (product by a constant, constant term, union of summations, decomposition, index translation and index reflection)
- Finally, we will compute some **important summations**: the sum of the first n natural numbers, of the first n odd numbers and of the first n even numbers

The summation symbol

Definition 1: summation

Let $a_1, a_2, \ldots, a_n$ be n real numbers. Their sum

$$a_1 + a_2 + \cdots + a_n$$

can be written in compact form with the summation symbol:

$$\sum_{k=1}^{n} a_k$$

which is read: "sum for k from 1 to n of a_k". The symbol k is called the summation index.

- The summation symbol is therefore just a shorthand, which is nevertheless very useful when the terms a_k are defined explicitly as a function of the index k.

Example

$$\sum_{k=1}^{10} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{k=3}^{n} k^2 = 3^2 + 4^2 + 5^2 + \cdots + n^2$$

Dummy index

- The summation index is a **dummy index**. This means that if we replace k with i, j or any other index (in all its occurrences), the value of the summation does not change.

Example

We have:

$$\sum_{k=1}^{n} k^2 \quad = \quad \sum_{i=1}^{n} i^2$$

since both symbols denote the sum of the squares of the first n natural numbers (without zero).

On the other hand, we have:

$$\sum_{k=1}^{n} k^2 \quad \neq \quad \sum_{k=1}^{m} k^2$$

since the two symbols denote the sum of, respectively, the first n or the first m squares (if $n \neq m$ the result will be different).

Main properties of summations

We now see six fundamental properties of summations, which will help us to manipulate and simplify expressions involving the symbol $\sum$.

Product by a constant

Observation 1: product by a constant

Given a summation $\sum_{k=1}^{n} a_k$ *and a real number* $c \in \mathbb{R}$, *we have:*

$$\sum_{k=1}^{n} (c \cdot a_k) = c \sum_{k=1}^{n} a_k$$

Proof

By the distributive property we have:

$$\underbrace{c\,a_1 + c\,a_2 + \cdots + c\,a_n}_{=\sum_{k=1}^{n}(c\cdot a_k)} = \underbrace{c\,(a_1 + a_2 + \cdots + a_n)}_{=c\,\sum_{k=1}^{n} a_k}$$

□

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Summation with a constant term

Observation 2: summation with a constant term

For every natural number $n \geq 1$ and every real number $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^{n} c = c \cdot n$$

Proof

We have:

$$\underbrace{c + c + \cdots + c}_{= \sum_{k=1}^{n} c \quad \text{that is, } c \text{ added } n \text{ times}} = c\, n$$

□

Union of summations

Observation 3: union of summations

Given two summations $\sum_{k=1}^{n} a_k$ *and* $\sum_{k=1}^{n} b_k$*, we have:*

$$\sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k = \sum_{k=1}^{n} (a_k + b_k)$$

Proof

We have:

$$\underbrace{a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n}_{=\sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k}$$

$$= \underbrace{a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n}_{=\sum_{k=1}^{n} (a_k + b_k)}$$

□

- The three properties that follow (decomposition, index translation and index reflection) are simply different ways of writing and/or ordering the terms of the summations
- To prove them it is enough to write out the terms of the two summations and observe that they are the same

Decomposition

Observation 4: decomposition

Given two natural numbers $n \geq 1$ and $m \geq 1$, we have:

$$\sum_{k=1}^{n+m} a_k = \sum_{k=1}^{n} a_k + \sum_{k=n+1}^{n+m} a_k$$

Proof

The terms of the summation $\sum_{k=1}^{n+m} a_k$ can be grouped into the first n and the last m ones:

$$\underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^{n} a_k} + \underbrace{a_{n+1} + a_{n+2} + \cdots + a_{n+m}}_{=\sum_{k=n+1}^{n+m} a_k}$$

$$= \underbrace{a_1 + a_2 + \cdots + a_n + a_{n+1} + \cdots + a_{n+m}}_{=\sum_{k=1}^{n+m} a_k}$$

□

Index translation

Observation 5: index translation

Given a summation $\sum_{k=1}^{n} a_k$ *and a natural number* $m \geq 1$*, we have:*

$$\sum_{k=1}^{n} a_k = \sum_{k=1+m}^{n+m} a_{k-m} = \sum_{k=1-m}^{n-m} a_{k+m}$$

Proof (1/2)

In the summation $\sum_{k=1+m}^{n+m} a_{k-m}$ the index k ranges from $1+m$ to $n+m$, and therefore the index $k-m$ ranges from 1 to n:

$$\underbrace{a_{(1+m)-m} + a_{(2+m)-m} + \cdots + a_{(n+m)-m}}_{=\sum_{k=1+m}^{n+m} a_{k-m}}$$

$$= \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^{n} a_k}$$

Proof (2/2)

In the same way, in the summation $\sum_{k=1-m}^{n-m} a_{k+m}$ the index k ranges from $1-m$ to $n-m$, and therefore the index $k+m$ ranges from 1 to n:

$$\underbrace{a_{(1-m)+m} + a_{(2-m)+m} + \cdots + a_{(n-m)+m}}_{=\sum_{k=1-m}^{n-m} a_{k+m}}$$

$$= \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^{n} a_k}$$

□

Index reflection

Observation 6: index reflection

Given a summation $\sum_{k=1}^{n} a_k$*, we have:*

$$\sum_{k=1}^{n} a_k = \sum_{k=1}^{n} a_{n-k+1} = \sum_{k=0}^{n-1} a_{n-k}$$

Proof (1/2)

In the summation $\sum_{k=1}^{n} a_{n-k+1}$ the index k ranges from 1 to n, and therefore the index $n-k+1$ ranges from n to 1, that is, the terms are the same ones listed in reverse order:

$$\underbrace{a_n + a_{n-1} + \cdots + a_1}_{=\sum_{k=1}^{n} a_{n-k+1}} = \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^{n} a_k}$$

Proof (2/2)

In the same way, in the summation $\sum_{k=0}^{n-1} a_{n-k}$ the index k ranges from 0 to $n-1$, and therefore the index $n-k$ ranges from n to 1:

$$\underbrace{a_n + a_{n-1} + \cdots + a_1}_{=\sum_{k=0}^{n-1} a_{n-k}} = \underbrace{a_1 + a_2 + \cdots + a_n}_{=\sum_{k=1}^{n} a_k}$$

□

Some important summations

Using the properties just seen, we now compute in closed form some important summations.

Sum of the first n natural numbers

Observation 7: sum of the first n natural numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^{n} k = \frac{n\,(n+1)}{2}$$

Proof

Using the index reflection:

$$\sum_{k=1}^{n} k = \frac{1}{2}\left(\sum_{k=1}^{n} k + \sum_{k=1}^{n} k\right) = \frac{1}{2}\left(\sum_{k=1}^{n} k + \sum_{k=1}^{n} (n-k+1)\right)$$

$$= \frac{1}{2}\sum_{k=1}^{n}(k+n-k+1) = \frac{1}{2}\sum_{k=1}^{n}(n+1) = \frac{n(n+1)}{2}$$

□

Sum of the first n odd numbers

Observation 8: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^{n}(2\,k-1) = n^2$$

or equivalently

$$\sum_{k=0}^{n-1}(2\,k+1) = n^2$$

Proof (1/2)

Using the properties of summations we have:

$$\sum_{k=1}^{n}(2\,k-1) = 2\sum_{k=1}^{n} k - \sum_{k=1}^{n} 1$$

$$= 2\left(\frac{n\,(n+1)}{2}\right) - n = n^2 + n - n = n^2$$

Proof (2/2)

For the second form:

$$\sum_{k=0}^{n-1}(2\,k+1) = 2\sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1$$

$$= 2\sum_{k=1}^{n}(k-1) + n = 2\left(\sum_{k=1}^{n} k - \sum_{k=1}^{n} 1\right) + n$$

$$= 2\left(\frac{n\,(n+1)}{2} - n\right) + n = n^2 + n - 2\,n + n = n^2$$

□

Sum of the first n even numbers

Observation 9: sum of the first n even numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^{n} 2\,k = n\,(n+1)$$

Proof

$$\sum_{k=1}^{n} 2\,k = 2\sum_{k=1}^{n} k = 2\left(\frac{n\,(n+1)}{2}\right) = n\,(n+1)$$

□

- The three important summations we have just seen can also be proved by induction, in the chapter "Principle of induction" of Part 1
- Here we have obtained them using only the properties of summations

Summary

Summations:

- $\sum_{k=1}^{n} a_k$ is a compact notation for $a_1 + a_2 + \cdots + a_n$; the index k is a **dummy index**
- The six properties: product by a constant, constant term, union of summations, decomposition, index translation and index reflection

Important summations ($n \geq 1$):

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} (2k-1) = n^2, \qquad \sum_{k=1}^{n} 2k = n(n+1)$$

Geometric progressions

Introduction: Geometric progressions

- In this section we introduce **geometric progressions**: sequences in which each term is obtained from the previous one by multiplying it by a constant, the **common ratio**
- We will derive the **closed formula of the k-th term** and the formula for the **sum of the first n terms** of a geometric progression
- This formula appears often in the course (for example in the study of geometric series)

Geometric progression

Definition 2: geometric progression

A sequence of real numbers is a ***geometric progression*** *if the ratio between each term (starting from the second one) and the previous one is constant. This constant is called the common ratio of the progression.*

Terms of a geometric progression

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, the associated geometric progression is:

$$a, \quad a\,q, \quad a\,q^2, \quad a\,q^3, \quad a\,q^4, \quad \dots$$

Each term (starting from the second one) is obtained from the previous one by multiplying it by q. The k-th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a\,q^{k-1}$ and we have:

$$\begin{aligned}
a\,q^{1-1} &= a\,q^0 = a && \text{first term, in position } k = 1 \\
a\,q^{2-1} &= a\,q^1 = a\,q && \text{second term, in position } k = 2 \\
a\,q^{3-1} &= a\,q^2 && \text{third term, in position } k = 3 \\
a\,q^{4-1} &= a\,q^3 && \text{fourth term, in position } k = 4 \\
&\dots && \dots
\end{aligned}$$

k-th term

Observation 10: k-th term of a geometric progression

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, let t_k denote the term in position k. The recursive relation that defines the progression is:

$$t_1 = a \qquad \text{and} \qquad t_k = t_{k-1}\, q \qquad \text{for every } k \geq 2$$

and the closed formula of the k-th term is:

$$t_k = a\, q^{k-1} \qquad \text{for every } k \geq 1$$

Proof

We apply repeatedly the recursive relation:

$$t_1 = a = a\, q^0$$

$$t_2 = t_1\, q = a\, q = a\, q^1$$

$$t_3 = t_2\, q = a\, q\, q = a\, q^2$$

$$t_4 = t_3\, q = a\, q^2\, q = a\, q^3$$

$$\vdots$$

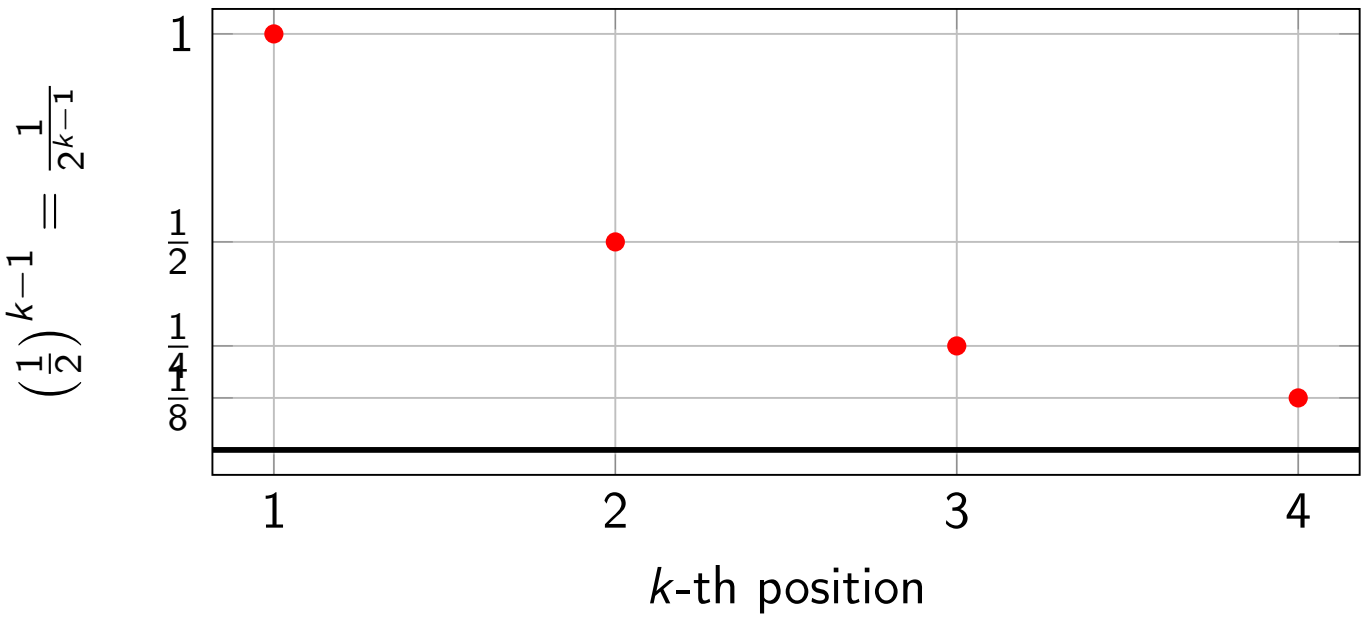
$$t_k = t_{k-1}\, q = a\, q^{k-2}\, q = a\, q^{k-1}$$

□

Example (1/3)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

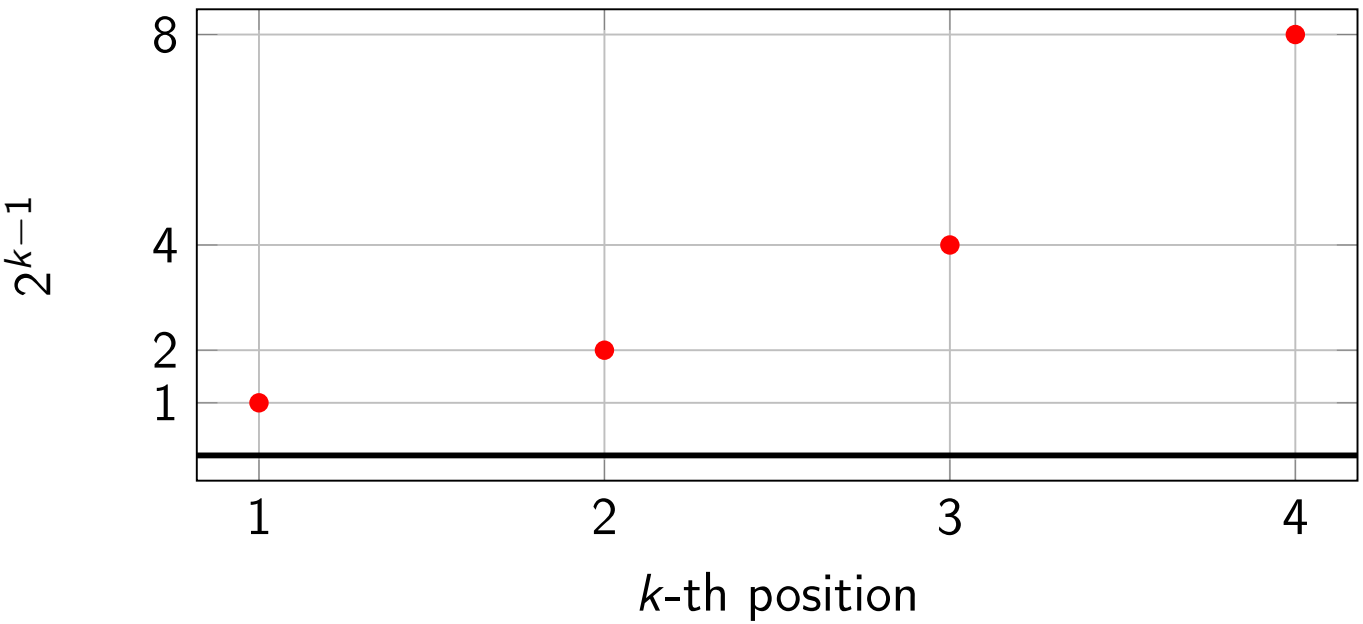
$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}$$



Example (2/3)

- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, \quad 2, \quad 4, \quad 8$$

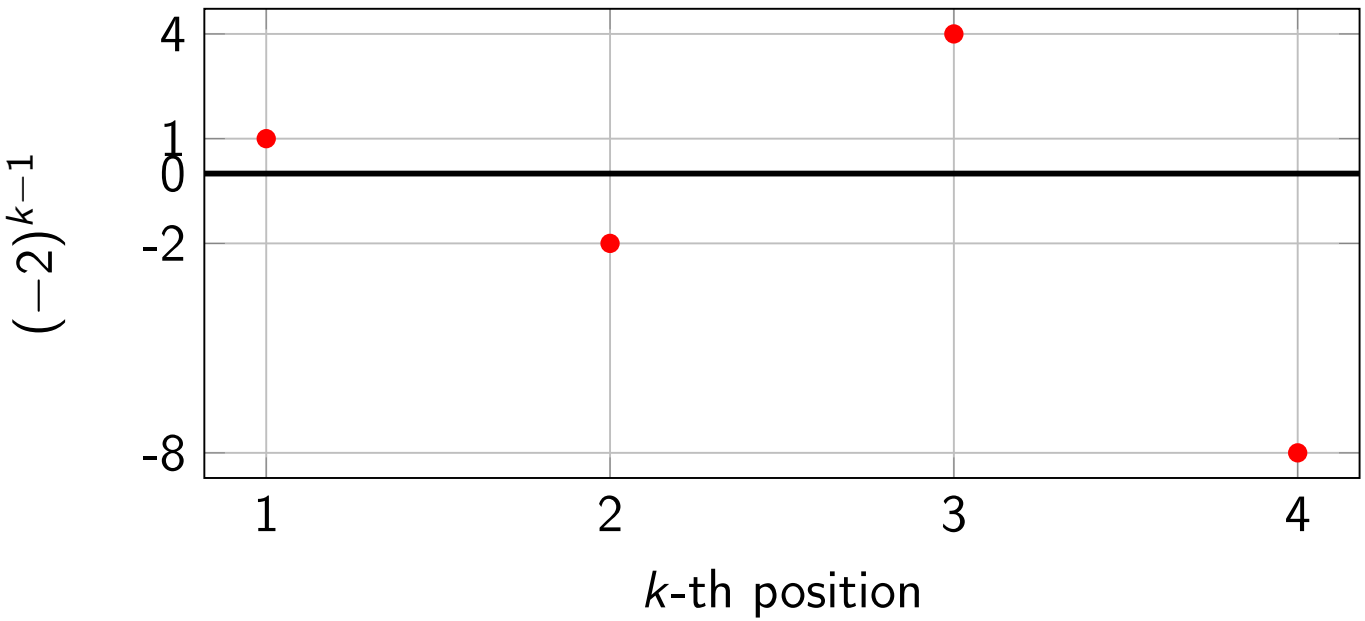


Example (3/3)

- With $a = 1$ and $q = -2$, the first 4 terms are:

$$1, \quad -2, \quad 4, \quad -8$$

- With a negative common ratio the terms change sign at every step



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Sums of the terms of geometric progressions

We now see how to compute in closed form the sum of the first n terms of a geometric progression.

Sum of the first n terms

Observation 11: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^{n} q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if} \quad q \neq 1 \\ n & \text{otherwise} \end{cases}$$

Proof (1/2)

If $q \neq 1$, we prove the equation in the following equivalent form:

$$(q-1)\sum_{k=1}^{n} q^{k-1} = q^n - 1$$

Applying the properties of summations, in particular the product by a constant and the index translation, we get:

$$(q-1)\sum_{k=1}^{n} q^{k-1} = q\sum_{k=1}^{n} q^{k-1} - \sum_{k=1}^{n} q^{k-1} =$$

$$= \sum_{k=1}^{n} q^{k} - \sum_{k=1}^{n} q^{k-1} = \sum_{k=1}^{n} q^{k} - \sum_{k=0}^{n-1} q^{k} =$$

$$= \sum_{k=1}^{n-1} q^{k} + q^n - \left(1 + \sum_{k=1}^{n-1} q^{k}\right) = q^n - 1$$

Proof (2/2)

Dividing by $q - 1 \neq 0$ we obtain:

$$\sum_{k=1}^{n} q^{k-1} = \frac{q^n - 1}{q - 1}$$

If $q = 1$, we have instead:

$$\sum_{k=1}^{n} q^{k-1} = \sum_{k=1}^{n} 1 = n$$

□

The common ratio can be negative or zero

- The proof never uses the sign of the common ratio: the formula holds for every $q \in \mathbb{R}$
- The only case to be made precise is $q = 0$: the first term is $q^{1-1} = 0^0$, which here is understood to be equal to 1 (convention $0^0 = 1$)
- With this convention, for every $n \geq 1$:

$$\sum_{k=1}^{n} 0^{k-1} = 1 + 0 + \cdots + 0 = 1 \qquad \text{and} \qquad \frac{0^n - 1}{0 - 1} = \frac{-1}{-1} = 1$$

Extension to the case $a \in \mathbb{R}$

- Given $q \in \mathbb{R}, n \in \mathbb{N}, n \geq 1$ and $a \in \mathbb{R}$, the formula extends as follows:

$$\sum_{k=1}^{n} a\, q^{k-1} = \begin{cases} a\, \left(\frac{q^n - 1}{q-1}\right) & \text{if} \quad q \neq 1 \\ a\, n & \text{otherwise} \end{cases}$$

since

$$\sum_{k=1}^{n} a\, q^{k-1} = a \sum_{k=1}^{n} q^{k-1}$$

Equivalent form

- Clearly:

$$\frac{q^n - 1}{q - 1} = \frac{1 - q^n}{1 - q}$$

and hence we also have:

$$\sum_{k=1}^{n} a\, q^{k-1} = \begin{cases} a\, \left(\frac{1-q^n}{1-q}\right) & \text{if} \quad q \neq 1 \\ a\, n & \text{otherwise} \end{cases}$$

- The formula for the sum of the first n terms can also be proved by induction, in the chapter "Principle of induction" of Part 1
- We now see three numerical examples, which check the formula with a direct computation

Example (1/4)

- With $a = 2$ and $q = \frac{1}{2}$, the first 4 terms are:

$$2, \quad 1, \quad \frac{1}{2}, \quad \frac{1}{4} \qquad \text{and} \qquad 2 + 1 + \frac{1}{2} + \frac{1}{4} = \frac{8 + 4 + 2 + 1}{4} = \frac{15}{4}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} 2 \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^{4} \frac{2}{2^{k-1}} = 2 \, \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}}$$

$$= 2 \, \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \, 4 = \frac{15}{4}$$

Example (2/4)

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8} \qquad \text{and} \qquad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8+4+2+1}{8} = \frac{15}{8}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^{4} \frac{1}{2^{k-1}} = \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16}\, 2 = \frac{15}{8}$$

Example (3/4)

- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, \quad 2, \quad 4, \quad 8 \qquad \text{and} \qquad 1 + 2 + 4 + 8 = 15$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} 2^{k-1} = \frac{2^4 - 1}{2 - 1} = 16 - 1 = 15$$

Example (4/4)

- With $a = 1$ and $q = -2$, the first 4 terms are:

$$1, \quad -2, \quad 4, \quad -8 \qquad \text{and} \qquad 1 - 2 + 4 - 8 = -5$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} (-2)^{k-1} = \frac{(-2)^4 - 1}{-2 - 1} = \frac{15}{-3} = -5$$

Summary

Geometric progressions:

- First term a, common ratio q: terms $a,\ aq,\ aq^2,\ \ldots$
- Recursive relation $t_1 = a,\ t_k = t_{k-1}\, q$; closed formula $t_k = a\, q^{k-1}$

Sum of the first n terms:

- If $q \neq 1$: $\displaystyle\sum_{k=1}^{n} a\, q^{k-1} = a\, \frac{q^n - 1}{q - 1} = a\, \frac{1 - q^n}{1 - q}$
- If $q = 1$: $\displaystyle\sum_{k=1}^{n} a\, q^{k-1} = a\, n$

Arithmetic progressions

Introduction: Arithmetic progressions

- In this section we introduce **arithmetic progressions**: sequences in which each term is obtained from the previous one by adding a constant, the **common difference**
- They are the additive analogue of geometric progressions: there one multiplies by q, here one adds d
- Here too we will derive the **closed formula of the k-th term** and the formula for the **sum of the first n terms**, which we will use in the study of series

Arithmetic progression

Definition 3: arithmetic progression

A sequence of real numbers is an ***arithmetic progression*** *if the difference between each term (starting from the second one) and the previous one is constant. This constant is called the common difference of the progression.*

Terms of an arithmetic progression

Given the first term $a \in \mathbb{R}$ and the common difference $d \in \mathbb{R}$, the associated arithmetic progression is:

$$a, \quad a + d, \quad a + 2\,d, \quad a + 3\,d, \quad a + 4\,d, \quad \dots$$

Each term (starting from the second one) is obtained from the previous one by adding d to it. The k-th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a + (k-1)\,d$ and we have:

$a + (1-1)\,d = a + 0\,d = a$	first term ($k = 1$)
$a + (2-1)\,d = a + 1\,d = a + d$	second term ($k = 2$)
$a + (3-1)\,d = a + 2\,d$	third term ($k = 3$)
$a + (4-1)\,d = a + 3\,d$	fourth term ($k = 4$)
$\dots$	$\dots$

k-th term

Observation 12: k-th term of an arithmetic progression

Given the first term $a \in \mathbb{R}$ and the common difference $d \in \mathbb{R}$, let t_k denote the term in position k. The recursive relation that defines the progression is:

$$t_1 = a \qquad \text{and} \qquad t_k = t_{k-1} + d \qquad \text{for every } k \geq 2$$

and the closed formula of the k-th term is:

$$t_k = a + (k-1)\, d \qquad \text{for every } k \geq 1$$

Proof

We apply repeatedly the recursive relation:

$$t_1 = a = a + 0\,d$$

$$t_2 = t_1 + d = a + d = a + 1\,d$$

$$t_3 = t_2 + d = a + d + d = a + 2\,d$$

$$t_4 = t_3 + d = a + 2\,d + d = a + 3\,d$$

$$\vdots$$

$$t_k = t_{k-1} + d = a + (k-2)\,d + d = a + (k-1)\,d$$

□

The same formula is obtained by a second argument, which does not use the dots: we add all the consecutive differences and the sum turns out to be **telescopic**.

Alternative proof (1/2)

We write the recursive relation in the form $t_j - t_{j-1} = d$ for every position j from 2 to k:

$$t_2 - t_1 = d$$

$$t_3 - t_2 = d$$

$$t_4 - t_3 = d$$

$$\vdots$$

$$t_k - t_{k-1} = d$$

We now add all these $k - 1$ equalities: on the right-hand side we add d to itself $k - 1$ times and we obtain $d\,(k - 1)$.

Alternative proof (2/2)

On the left-hand side the sum is **telescopic**: every term appears twice, once with the sign $+$ and once with the sign $-$, except t_k and t_1:

$$\underbrace{(t_2 - t_1) + (t_3 - t_2) + \cdots + (t_k - t_{k-1})}_{= \sum_{j=2}^{k} (t_j - t_{j-1})} = t_k - t_1 = t_k - a$$

Equating the two sides we get $t_k - a = d\,(k-1)$, that is:

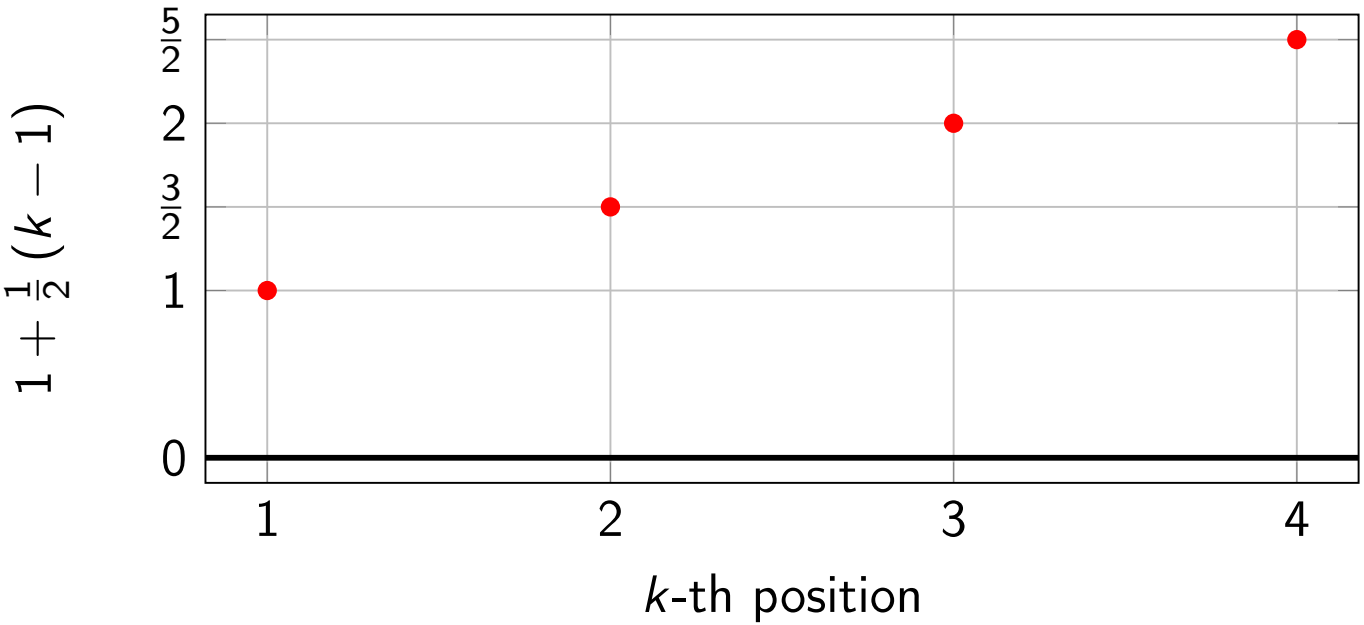
$$t_k = a + (k-1)\,d \qquad \text{for every } k \geq 1$$



Example (1/2)

- With $a = 1$ and $d = \frac{1}{2}$, the first 4 terms are:

$$1, \quad \frac{3}{2}, \quad 2, \quad \frac{5}{2}$$

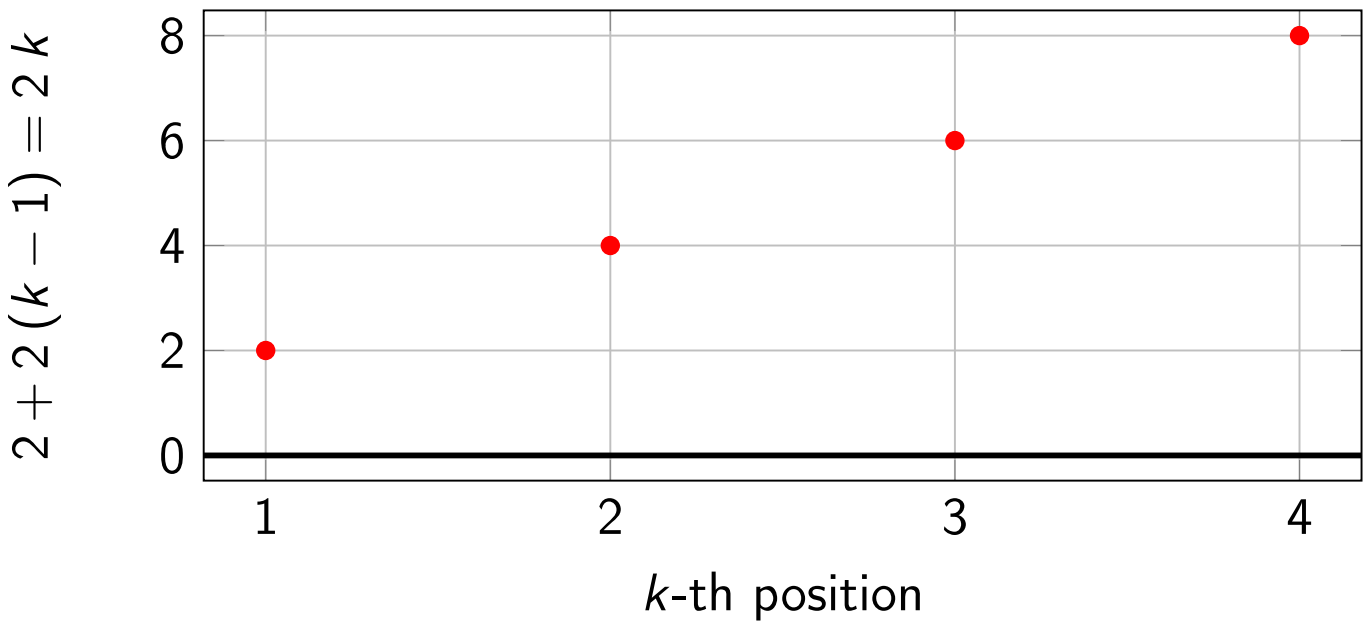


Example (2/2)

- With $a = 2$ and $d = 2$, the first 4 terms are:

$$2, \quad 4, \quad 6, \quad 8$$

that is, the first 4 even numbers (without zero)



Sums of the terms of arithmetic progressions

We now see how to compute in closed form the sum of the first n terms of an arithmetic progression: we use the index reflection, the same technique as for the sum of the first n natural numbers.

Sum of the first n terms

Observation 13: sum of the first n terms of the arithmetic progression ($a = 0$)

Given $d \in \mathbb{R}$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^{n} d\,(k-1) = d\,\left(\frac{n\,(n-1)}{2}\right)$$

Proof

Using the product by a constant and the index reflection:

$$\sum_{k=1}^{n} d\,(k-1) = \frac{d}{2}\left(\sum_{k=1}^{n}(k-1) + \sum_{k=1}^{n}(k-1)\right)$$

$$= \frac{d}{2}\left(\sum_{k=1}^{n}(k-1) + \sum_{k=1}^{n}(n-k)\right)$$

$$= \frac{d}{2}\sum_{k=1}^{n}(k-1+n-k) = \frac{d}{2}\sum_{k=1}^{n}(n-1)$$

$$= d\left(\frac{n\,(n-1)}{2}\right)$$

(the term in position k is $a_k = k-1$, hence $a_{n-k+1} = n-k$) □

Extension to the case $a \in \mathbb{R}$

- Given $d \in \mathbb{R}, n \in \mathbb{N}, n \geq 1$ and $a \in \mathbb{R}$, the formula extends as follows:

$$\sum_{k=1}^{n} \left(a + d\,(k-1)\right) = a\,n + d\,\left(\frac{n\,(n-1)}{2}\right)$$

since

$$\sum_{k=1}^{n} \left(a + d\,(k-1)\right) = \underbrace{\sum_{k=1}^{n} a}_{=a\,n} + \sum_{k=1}^{n} d\,(k-1)$$

Equivalent form

- Reflecting the index, the term in position k and the one in position $n-k+1$ have constant sum $t_k + t_{n-k+1} = t_1 + t_n$, and therefore:

$$\sum_{k=1}^{n} \left(a+d(k-1)\right) = \frac{n}{2}\left(\underbrace{a}_{=t_1} + \underbrace{a+(n-1)\,d}_{=t_n}\right) = n\left(\frac{t_1+t_n}{2}\right)$$

- The sum of the first n terms is therefore n times the average of the first and the last term

Special cases already known

- With $a = 1$ and $d = 1$ the progression is $1, 2, 3, \ldots$ and the formula gives

$$n + \frac{n\,(n-1)}{2} = \frac{n\,(n+1)}{2}$$

that is, the sum of the first n natural numbers (without zero)

- With $a = 1$ and $d = 2$ the progression is $1, 3, 5, \ldots$ and the formula gives

$$n + 2\,\frac{n\,(n-1)}{2} = n^2$$

that is, the sum of the first n odd numbers

Example (1/2)

- With $a = 1$ and $d = \frac{1}{2}$, the first 4 terms are:

$$1, \quad \frac{3}{2}, \quad 2, \quad \frac{5}{2} \qquad \text{and} \qquad 1 + \frac{3}{2} + 2 + \frac{5}{2} = \frac{14}{2} = 7$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} \left(1 + \frac{1}{2}(k-1)\right) = 1 \cdot 4 + \frac{1}{2}\left(\frac{4 \cdot 3}{2}\right) = 4 + 3 = 7$$

Example (2/2)

- With $a = 2$ and $d = 2$, the first 4 terms are:

$$2, \quad 4, \quad 6, \quad 8 \qquad \text{and} \qquad 2 + 4 + 6 + 8 = 20$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^{4} \left(2 + 2\left(k - 1\right)\right) = 2 \cdot 4 + 2 \left(\frac{4 \cdot 3}{2}\right) = 8 + 12 = 20$$

and with the equivalent form: $4 \left(\frac{2+8}{2}\right) = 20$

Summary

Arithmetic progressions:

- First term a, common difference d: terms $a,\ a+d,\ a+2d,\ \ldots$
- Recursive relation $t_1 = a,\ t_k = t_{k-1} + d$; closed formula $t_k = a + (k-1)\,d$

Sum of the first n terms:

- $\displaystyle\sum_{k=1}^{n} \big(a + d\,(k-1)\big) = a\,n + d\,\frac{n(n-1)}{2} = n\,\frac{t_1 + t_n}{2}$
- Special cases: sum of the first n natural numbers ($a = d = 1$) and of the first n odd numbers ($a = 1,\ d = 2$)

Final summary

- The symbol $\sum$ allows us to write sums of many terms in compact form; its index is a dummy index
- The six properties of summations (product by a constant, constant term, union, decomposition, index translation and index reflection) allow us to compute sums in closed form
- Important summations:

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n}(2k-1) = n^2, \qquad \sum_{k=1}^{n} 2k = n(n+1)$$

- In a geometric progression with common ratio $q \neq 1$:

$$\sum_{k=1}^{n} a\, q^{k-1} = a\, \frac{q^n - 1}{q - 1}$$

- In an arithmetic progression:

$$\sum_{k=1}^{n} \big(a + d\,(k-1)\big) = a\, n + d\, \frac{n(n-1)}{2}$$